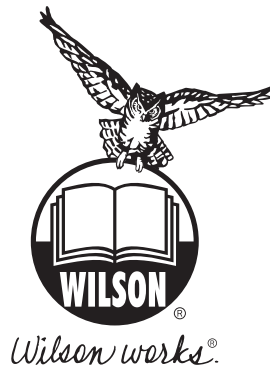


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The Bent Pyramid

The mysterious pyramids of Egypt have attracted visitors for thousands of years. Early kings of Egypt built a type of pyramid with bench-shaped piles stacked like rising steps. Later, smooth-sided pyramids were built.

Among the many pyramids of Egypt, an unusual one stands out. It's called the Bent Pyramid. This pyramid does classify as a step pyramid. However, it appears as if the bottom part is shedding. The sides of the pyramid start out at a 54 degree angle. Yet, toward the top, the slope drops to 43 degrees. This gives the pyramid a bent appearance. Although it remains as somewhat of a mystery, it's possible the pyramid was unstable during construction. At some point, the builders may have had to make an adjustment to avoid a horrifying crumbling of the pyramid.

The Bent Pyramid was built by Pharaoh Snefru more than 4,000 years ago. This pyramid is considered an important part of the progression from step style to smooth-sided pyramid construction. In fact, Snefru is credited with experimenting with different construction approaches. This experimentation resulted in what some imply are the first true pyramids. Snefru's son, Khufu, followed him as ruler of Egypt. He took what he learned from his father and built the Great Pyramid of Giza. This is possibly the most well known of all pyramids.

Although he lived a very long time, researchers have not been able to specify that the Bent Pyramid was built for Snefru to occupy after dying. Still, the Bent Pyramid and others like it have served to identify the names and symbolize the tales of Egypt's ancient kings to this day.

Wild Cats of North America

Six types of wild cats occupy North America.

However, only three are found commonly in Canada and the United States. They are the bobcat, the Canada lynx, and the puma.

The bobcat is considered a mid-sized cat and it's found throughout the United States, Mexico, and southern Canada. In fact, it's no mystery that they are the most common of the wild cats in this region. They are named after their short "bobbed" tails. The bobcat has fur growing from the tips of its ears and has clumps of fur sticking out from its cheeks. A bobcat has rather short legs and a spotted, sometimes colorful, coat.

Canada lynxes live in Canada, Alaska, and the northern U.S. They have long legs and wide paws that are covered in hair which help them walk across the

snow. A lynx's coat lacks spots and is usually colored pale brown or gray. One can also distinguish between a bobcat and a Canada lynx by looking carefully at the tail. A lynx has an all black tail while bobcats' tails have black bars and white fur on the bottom.

The puma is a large wild cat found over an extensive area of North America - from the Yukon up in Canada all the way down to southern Chile. Despite its large size, a puma has a mysteriously small head. Pumas also have rounded ears and a coat with no markings. Pumas have long tails with a black tip and they can range in color from dark brown to gray.

Bobcats, Canada lynxes, and pumas are all wild cats of North American, but they have distinct traits that set them apart from one another.

The Sense of Touch

When we hold a glass and pour water into it, we feel it getting heavy. We apply our skillful senses to analyze and decide when we should stop pouring. Oxygen may prevent us from dying, but our senses help guide us through daily life.

Paralysis is an inability to move or feel sensation. When someone is paralyzed, losing their sense of touch greatly impacts everyday life. Doctors can supply some assistance. However, most often there is no cure. Despite this fact, scientists work hard to deconstruct the mysteriousness that surrounds the topic of paralysis.

Researchers have even placed sensors in a paralyzed person's head. These sensors were then connected to a robotic arm. The sensation the prosthetic arm offered

was not exactly the same as a regular arm. Yet, the test subject reported that it felt lifelike enough. As a result, he dropped fewer things, was able to identify changes through touch, and he performed tasks more quickly.

Prosthetic and robotic arms are very useful on their own. They can do much of what a normal arm does. However, their users are relying on other senses to make up for their loss of touch. This often forces major lifestyle changes. With the sensors placed on the brain, researchers try to help people regain a sense of touch. Yet, this is just part of the equation. These advances offer much hope to those who have lost the ability to move or feel parts of their bodies.

Talented Mimics

Listening to a lyrebird's symphony is an extraordinary experience. These mysterious birds have an extensive vocal range. In addition, they are able to replicate and amplify sound. It's said to be hypnotizing.

Though shy and reclusive, we are beginning to solve the mystery of the lyrebird and its song. Simply put, they sing to mark their territory and attract mates. However, there is more to lyrebirds than just their voices. In fact, there are two types of lyrebirds - the superb lyrebird and the Albert's lyrebird. Both can be found in and around Australia. While both are able to replicate the sounds of other animals, humans, and even machinery, these birds are quite distinctive from each other.

The Albert's lyrebird is smaller than the superb lyrebird. It also has a more muted coloring. In contrast,

the superb lyrebird has vibrant coloring with large plumage. In fact, this lyrebird was named as such because of its superb feathers. You see, when they are lying out straight, they resemble a lyre, an old-style musical instrument.

The population of the superb lyrebird is much larger than that of Albert's lyrebird. The superb lyrebird acts as an engineer of sorts for its local ecosystem. It digs for food, modifying and shifting sizable amounts of soil. This helps diversify the microorganism population and encourages seed growth.

Whether it is mimicking other living beings, syncing up its sound to surrounding machinery, or exemplifying a habitat engineer, the lyrebird's hype is easy to justify. This is one impressive and talented mimic!

Return of the Nene PART ONE

The journey of the nene, or Hawaiian goose, has been quite the story. Hawaii's state bird is a frequent flier on the endangered species list. After an undeniable population drop in 1960, worry surrounding the bird increased. Something had to be done to save them.

Difficulties for the bird began in the late 1700s. The arrival of new travelers was problematic. The nene population was variable and never as plentiful as many other types of geese. Still, experts estimate that there were as many as 25,000 living in Hawaii then. Hunting quickly reduced these healthier numbers.

More and more travelers came over the years. Suddenly, the food these geese relied on was disappearing. They did not have enough to eat which reduced their numbers. The voyagers also brought other

animals with them. This included rats, dogs, and cats. Researchers identified the problem. The geese were competing with these animals for food. Their population further decreased.

Mongoose did even more damage to the nene. The mongoose, a small ferretlike mammal, was swiftly introduced in 1883 to deal with rats in crops. Too bad this fix was not researched beforehand. As expected, the mongoose families stayed busy snacking on rats. Sadly, they also satisfied their hunger with the eggs of grounded birds, like the nene.

Between unmerciful mongooses eating their eggs and unreliable resources, the nene were nearly wiped out. With no protection policies in place, their population hit an all-time low of 30 in 1960.

Return of the Nene PART TWO

There are many animals classified as at risk in the U.S. Sadly, few conservation agencies were active locally before the mid-1960s. In fact, the first federal list of endangered species was not released until 1967. A lot of animals were identified on this first list, including the nene. However, it would have been unwise to wait this long to act. With the population dropping to just 30, many worried it was already too late to save the nene. Thankfully, these worries did not stop protection efforts. Conservationists hurried into action.

Efforts began primarily with state and private support. Hatcheries were built in the early 1960s. The geese were studied as experts tried to keep them from dying out. In 1962, many geese were brought back into the wild. They were carried down to a pen and placed

there temporarily to adjust to their surroundings before release.

The nene breeding program was successful. When the federal list of endangered species was released in 1967, more policies were put in place. Guarded boundaries were established. There were also sturdier punishments for those interfering with or injuring protected wildlife. It became easier to protect endangered animals. Finally, the nene numbers steadily increased.

Today, there are around 2,500 nene in the wild. These numbers allowed the nene's conservation classification to change from endangered to vulnerable, or at risk. Busy conservation agencies still keep an eye on the nene. However, it appears these geese have made their way out of the woods.

Productive Plants

Treating maladies can be a tricky business. Many strategies for treating diseases and autoimmune disorders have undeniable drawbacks. From cost prohibitive medication to worry surrounding allergies, effective treatment may be difficult to administer. This is where botanists can help. Plant-based medications and support can be a safe, cost-effective alternative.

One of the biggest worries for those seeking medical treatments is the safety of the medication. Those with allergies to common additives often face justifiable difficulties complying with medical recommendations. With the help of plants like lettuce (yes, really!), certified alternative methods of delivery are possible. Vegetables are easily grown and readily available tools for effective delivery. With this modified treatment method, many

vaccines can be administered with just proteins, rather than the pathogen itself.

Some medications are difficult to store and expensive to produce. This means many people do not have access to these invaluable commodities. By using new modifiable delivery methods, many new medications have become shelf stable. This not only makes it easier to get medication to remote regions, it also reduces transportation and storage costs. The medication can be placed in pantries instead of fridges. This is way less expensive!

These modified plant-based alternatives will not replace all existing medications and deliveries. However, their addition to the mix could drastically improve accessibility and overall health.

The History of Money

What is money, exactly? Money is a currency.

Currencies are items exchanged between sellers and buyers with some specified value.

The first currencies were used thousands of years ago. Metal objects were introduced around 5000 BC. By 700 BC, coins were being used. These coins would represent something, like a pile of grain. They could be exchanged for a commodity, like a cow. This system only worked if the promise of grain was reliable. The cow seller had to worry about how and when he'd get his grain, as he couldn't verify the cow purchaser's integrity. If the grain wasn't there, the reliance on a promise didn't work well, and the cow owner would be justifiably angry.

Other groups throughout history identified their own currencies as beads, shells, or animal furs. These items

had their own value and did not represent bounties, or payments, elsewhere. Commodities exchanged with this system were more reliable, so people were more satisfied and less worried about the fairness of the exchange.

The earliest form of American currency was likely the wampum, small beads strung together. The first United States paper money became available in 1862.

Today, governments control their own certified currencies, coins, and paper. Each government sets up policies and strategies for how its currency is used. When countries want to exchange currencies, they must agree on the specified value of each other's money. Today's system has many complexities, such as mobile transactions, that keep modern businesses and bankers busy.

Dead Zones

Dead zones are sections of bodies of water where there are no plants or animals. The unsightly, algae-filled areas are unable to support life.

Dead zones are usually the result of pollution. High amounts of chemicals flood into the water and make too much algae. When combined with a lot of sunlight, this pollution can cause an overgrowth of algae, known as blooms. Algae blooms have a frightening impact on the water. First, the algae pulls in tightly, forming a wall that prevents light from entering. Without sunlight, much of the plant life is destroyed. Additionally, when the algae dies, it uses up oxygen in the water. As oxygen levels drop and daylight disappears, fish must fight to survive. The end result is a dead zone.

Plants and animals in these dead zones rarely make it out. Those fish that can get away

rightfully move to neighboring areas. Most are lost. Tragically, these dead zones can have even weightier consequences.

If an algae bloom grows large enough, it can actually make its own pollution. When this happens, it is called a harmful algae bloom. These algae blooms can increase the size of dead zones. They can happen in almost any type of water, too. This means highly used drinking water sources might be impacted if they are near these dead zones. The result is sick people and animals.

Not only do dead zones harm life in the water, they can have an equal impact on life around it. As pollution levels grow, more dead zones are cropping up, threatening the health and well-being of all life in and around them.

The Lighthouse Keeper

Richard's days begin before daylight and today is no exception. As a lighthouse keeper, Richard has many responsibilities. He has to make sure the light is operating correctly at all times. He must keep up with repairs and, rightly so, respond to emergencies such as shipwrecks.

Today, Richard is going to be busier than ever. There is a storm in sight. This means that the beacon, or light, needs to be operational for 24 hours a day until the storm passes. The bright light must be visible to ships during the height of the storm.

First, Richard goes to the bureau and gets his official lighthouse service uniform. The navy blue coat, vest, pants, and cap were never required until last year. Now, all lighthouse keepers needed to wear the uniform

with the letter K visible on the lapel, or collar, of the coat. Dressed for work and with his family safe and sound, Richard starts his day.

Most of the time Richard would do things like light the lamps, as well as clean the lenses and windows of the lighthouse, but not today. As the early morning fog rolls in, Richard begins the fog signal. Richard then makes his way up the winding staircase. High off the ground, he hears the winds begin to howl. Dark clouds and lightning approach and soon enough, mighty waves crash into the rocks threatening to flood the island's buildings. Because of his twenty-eight years of experience as a sailor and lighthouse keeper, Richard is not frightened, but he knows he has a long night ahead of him.

Spotlight on Right Whales

The North Atlantic right whale is one of the most endangered large animals. Weighing up to 75 tons, these whales got their name from being the “right” whales to hunt because they moved slowly and floated when they were killed. Overfishing by the whaling industry led to their protection in 1935. Even with these protections, humans still pose the greatest peril to right whales. As whales move through the water feeding on algae, collisions with freighters and entanglements in fishing gear cause injury and death. An increase in ocean noise levels is also a concern. Many conservation bureaus are working with fishing and shipping industries to find solutions to the steadily declining right whale population.

Vessel speed limits have been established during certain times of the year. The likelihood of fatal injuries

increases when ships carrying heavy freight collide with a whale while speeding. Slower vessel speeds also increase the crew's sight and lower the chances of collision.

To improve the plight of right whales, conservation groups are also advocating changes to fishing regulations. For example, “fixed gear,” such as lines attached to traps, cause severe injuries when whales get tightly entangled in them. Closing fixed gear fisheries seasonally at the height of whale activity might help solve this dilemma.

Questions remain about how to protect North Atlantic right whales while taking the interests of fishing and shipping industries into account. Time will tell whether the benefits to right whales outweigh the disadvantages imposed on these industries.

Disappearing Lakes

A huge lake vanished from East Antarctica nearly overnight.

The heavily weighted lake's strange vanishing act was discovered by chance. Insightful researchers caught a glimpse of a collapsed surface while studying satellite images of neighboring land masses. The frightening sight rightfully raised concerns about what might have happened to the missing body of water.

After their discovery, researchers backtracked satellite footage to gain some insight into the vanishing act. The images were closely studied. Research teams watched in astonishment as the height of the lake plummeted and a flood of water dropped into the ocean below. The lake took just three days to drain.

As it turned out, this flood was caused by the weight of the lake itself. The water became so heavy that it cracked the ice shelf under the lake. This is called a hydrofracture. This sudden and rapid flooding caused the water levels nearby to temporarily raise by one hundred and eighteen feet.

The drained lake has already begun to refill. However, the implications of hydrofracturing should not be taken lightly. There is a high likelihood hydrofracturing is increasing because of climate change, which is something researchers would like to spotlight. Because hydrofracturing is so unpredictable, its long-term implications with regard to sea levels is still a mystery. The lasting impacts of this collapse were slight, but as the former lake regains weight, another collapse seems inevitable.

How We Changed Dogs

Humans achieved their canine-related goals over thousands of years, successfully domesticating wolves to make the helpers and friends we now call dogs. However, this foreign interference has yielded some weird results. Not only did we change the way dogs look, but researchers believe we have changed the structure of their brains.

Research began with a series of brain scans, or images of the brain, from several different types of dogs. From a brief review of the scans, the research team quickly perceived a pattern. Although the shapes and sizes of the dog's heads varied considerably, six regions of the brain seemed to have the same variances. The scientists put the pieces together. They believed these regions were directly related to the years spent

breeding for specific traits. For example, beagles can smell cancer and collies can herd field animals.

It turned out that each of these six regions related to at least one breed-specific trait. Traditional police dog breeds had differences in the region that connected sight and smell. Dogs that were bred for protection showed differences in the region connected to stress and anxiety. Humans changed the structure of dogs' brains.

So what does this mean for man's best friend? Researchers hope it means more consideration when it comes to breeding and care. Humans wield a lot of power when it comes to our furry friends. Every action taken, for good or bad, can have an extraordinary and long-lasting impact.

The West Indies

The West Indies encompass beautiful islands and beautiful people. The large chain of islands is located in the Atlantic Ocean and the Caribbean Sea.

The Bahamas alone are comprised of approximately 3,000 islands. The waters surrounding these islands are crystal clear. Explorers, pirates, and merchants all landed on the shores of these islands. Settlers introduced slavery to the area until that was made illegal. Thus, Africa's impact on the culture of the Bahamas is still fiercely evident today.

The Greater Antilles is a series of large islands in the Caribbean Sea. Cuba is the largest of the islands and is filled with rivers and hills. South of Cuba, you will find Jamaica and its beautiful mountains and seaside cliffs.

For more than a hundred years, the Spanish chiefly controlled the islands of the Greater Antilles. However, the English, Dutch, and French seized control over some. Today, most of the islands are independent nations. They have become prime locations for foreign tourists to visit.

The Lesser Antilles is actually made up of two groups of islands - the Leeward Islands to the north and the Windward Islands to the south. The Windward Islands provide premier views from mountaintops. All of the Lesser Antilles islands have a sunny, mild climate. This is great for tourism and for growing fields of bananas, cotton, sugar, and spices.

The people of the West Indies have a rich history with special traditions and beliefs. Today, more than 44 million people call these islands their home.

Living with Prairie Dogs

The grass-covered, treeless plains of North America are home to many animals, including the prairie dog. Believe it or not, the prairie dog is not a dog at all. It's a rodent, a type of ground squirrel, that got its name because of the shrieking bark-like noises it makes.

Despite their adorable appearance, prairie dogs cause farmers much anxiety. Their fiendish tunneling and appetite for plants can destroy entire crops. Cattle looking to graze often find themselves fiercely competing with prairie dogs. Homeowners also become anxious when prairie dogs besiege their yards. The rodents can leave a garden looking like a battlefield.

The unsightly mounds of dirt left behind from burrowing prairie dogs, as well as the destruction of crops and gardens, leave many people wishing to

eliminate these pests forever. However, understanding the benefits of these neighbors could conceivably change their point of view. For example, in a world without prairie dogs, the mountain plover songbird would have no mounds of dirt to build their nests and the endangered black-footed ferret would lose its main source of food.

Solutions are being discussed across the plains because achieving a way to coexist means answering the following kinds of questions: Can prairie dogs be successfully relocated to areas where they are needed as a food source for ferrets, owls, and hawks? Can modifying the habitat with yielding barricades such as fences or tall grasses, shrubs, and trees help provide relief to homeowners and farmers?

The Film Festival Gig

Maddie and Zena began working for their university's newspaper at the start of their first year. The newspaper had a tiered selection process based on hard work. Dedicated staff were repaid for their efforts by getting the best assignments.

As newbies, Maddie and Zena were determined to prove themselves. They volunteered to work late, took on extra fieldwork, and even retrieved the unbelievably heavy photography equipment from storage. They seized every conceivable opportunity to improve their writing. By April, the pair had put in more newspaper hours than half the juniors and seniors on staff.

At the April meeting, the editor in chief began handing out assignments. The campus film festival series was beginning soon. Film students from all over

submitted their movies for review. It was the most prestigious writing gig, a crowning achievement for the hardest working senior. Still, Maddie and Zena believed they would receive great end-of-year gigs. As name after name was read off, the pair grew disappointed. Neither Maddie nor Zena was called.

Finally, the film festival was discussed. A gifted and hardworking senior was offered the gig. Then, the editor in chief made a surprising announcement. The film festival had grown too large. It would be impossible for one person to see and write about every movie. So this year, they would have two more dedicated staff members attend the festival and review the foreign film submissions. Maddie and Zena stared at each other in disbelief as the editor called their names.

The Kiwi Bird

If you visit New Zealand and spot a tiny brown bird with hair-like feathers, you might be admiring the kiwi bird. This bird gets its name from the sound of its call, “kee-wee, kee-wee.” The Latin species name is *Apteryx*, which means wingless and they are similar to other flightless birds such as the emu and ostrich.

Though they can’t fly, the kiwi bird has inferior wings measuring only about one inch. Each wing has a tiny weird claw on the end that biologists can’t explain. Because the kiwi bird spends its time on the ground, it has muscular legs that make up around a third of its total body weight.

There are five types of kiwi birds. The northern kiwi is the largest and it can reach a height of 20 inches and weigh about 4 pounds. They are often equal in size

to chickens. However, this is about all that it has in common with other birds.

The kiwi has a very long and odd-looking beak with whiskers on the exterior. The beak also has conveniently placed nostrils on the end. These brilliant characteristics combined with poor eyesight make most kiwis nocturnal, or active at night. Appropriately, they spend their days sleeping in hollow logs, dense vegetation, or holes in the ground. At night, their behaviors include enthusiastically tapping the ground to search for worms, bugs, berries, and seeds.

Kiwis produce weighty eggs for their size. This is because they hatch young that are mostly able to care for themselves. Because these traits are highly peculiar for birds, the kiwi has rightfully earned the curious title of “honorary mammal.”

Sunrise on the Summit

Ren and Cho had spent their whole lives looking at giant Mount Fuji. Yet, they had never climbed Japan's tallest mountain. When the siblings turned seventeen, they decided it was time to change that!

Growing up in the city of Gotemba allowed for a million brilliant views of this mysterious and symbolic mountain. Ren and Cho dreamed many times about the day they would make the climb up the familiar prism. To prepare, they had become experienced hikers, spending many weekends on the trails near their hometown.

When the day finally arrived to tackle Mount Fuji, the companions made sure they were ready for the 7-10 hour hike to the summit, or top, of the mountain. This was no inferior task as the diamond-shaped Mount Fuji reaches a height of 12,388 feet.

They each made sure their backpack had gloves, rain gear, a jacket, and a hat. Ren gathered the snacks, towels, and hiking sticks. Cho packed the water, headlamps, and spare batteries. Their climb would begin in the afternoon and continue throughout the evening.

Triumph! Their training paid off! Ren and Cho made it to the summit with relative ease, making their way across steep and rocky exterior parts of the trail without much difficulty. Even as the oxygen in the air decreased, they were able to keep up a satisfying pace. The trail station system provided spots to rest and re-energize.

Their timing was perfect! At approximately 6 a.m., Ren and Cho observed the most extraordinary sunrise they had even seen. Japan's nickname, The Land of the Rising Sun, was no longer a mystery to them!

Jonas Salk, American Virologist PART ONE

Born in 1914, Jonas Salk grew up poor in New York City. As a boy, Jonas was thin, small, and quite inferior at sports. His parents encouraged young Jonas to take his education seriously and he was obedient. After graduating from high school, Jonas attended the City College of New York. He earned a bachelor's degree in science. In 1939, Jonas earned his medical degree from New York University. After completing his internship, he earned a fellowship to the University of Michigan where he studied flu viruses.

In 1947, Dr. Salk took a position at the University of Pittsburgh. There he began studying polio, also called infantile paralysis. Polio is a disease that attacks the brain and spinal cord, most often in young

children. It is caused by a virus, not bacteria. Like the common cold, it can spread by a simple sneeze. Polio can cause paralysis of the arms and legs. Our thirty-second president, Franklin D. Roosevelt, had polio which resulted in him using a wheelchair to get around for much of his life.

Dr. Jonas Salk's genius and hard work resulted in the development of a vaccine for the crippling disease. National testing began in 1952 and continued for the next two years. It was one of the largest clinical trials in medical history. Approximately 1.8 million children who met the criteria were administered the vaccine during this test phase. In 1953, Dr. Salk even gave it to himself, his wife, and his children.

Jonas Salk, American Virologist PART TWO

In 1955, the polio vaccine was approved for general use. Dr. Salk did not get a patent on his discovery. He felt there should be no ownership criteria. Everyone who needed it should get it. Salk was soon considered a national hero. President Eisenhower even gave him a special citation during a ceremony held in the Rose Garden at the White House.

Within the first few years of use, the number of new polio cases plummeted in the United States. This vaccine was ultimately replaced with one developed by Albert Sabin because it was less expensive and easier to use.

Salk founded his own research organization called the Salk Center for Biological Studies in 1963.

Scientists working at the center focused their efforts on diseases such as multiple sclerosis and cancer. Salk served as the center's director until 1975 and continued researching serious illnesses late into his career. Of this he said, "The reward for work well done is the opportunity to do more."

In 1988, a global union of people formed with a mission to eradicate polio worldwide. Since then, more than 2.5 billion children have been immunized against polio thanks to the cooperation of more than 200 countries and 20 million volunteers.

Dr. Jonas Salk died on June 23, 1995 at his home in California. He will always be remembered as a champion of children and the man who stopped polio.